

Topology First-Year Exam, January 2023, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that a connected metric space having more than one point is uncountable.
2. Give definition of retraction. Show that a retraction is a quotient map.
3. Is the set of rationals $\mathbb{Q}$ in the subspace topology locally compact?
4. Show that the set of irrational numbers in $\mathbb{R}$ is a Baire space.
5. Let $f : X \rightarrow X$ be a map of a compact metric space (X, d) to itself that satisfies the following condition $d(f(x), f(y)) < d(x, y)$ for all $x \neq y$. Prove that f is continuous. Show that f has a unique fixed point.

Answer the following with complete definitions or statements or short proofs.

6. State the Intermediate Value Theorem.
7. Show that any continuous map $f : S^2 \rightarrow (0, 1)$ of the 2-sphere S^2 to the interval $(0, 1)$ is neither surjective nor injective.
8. State the Cantor–Schröder–Bernstein Theorem.
9. Is the space $\mathbb{R}_\ell$ connected? path connected?
10. What is a basis of a topology? Does the set $\{(a, b] \cup [c, d) \mid a < b < c < d\}$ form a basis of a topology on $\mathbb{R}$?